THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of 4^3 is: 1

(a) 68 **(b)** 12 **(c)** 64 **(d)** 48

2. The value of 2^{-4} is: 1

(a) 16 **(b)** $\frac{1}{16}$ **(c)** $\frac{1}{8}$ **(d)** -16

3. The number 5 in binary is: 1

(a) 110 **(b)** 101 **(c)** 111 **(d)** 100

4. A rhombus has each side 6 cm. Its perimeter is: 1

(a) 12 cm **(b)** 18 cm **(c)** 24 cm **(d)** 36 cm

5. The sum of the digits of 2748 is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 20

6. $\sqrt[3]{64}$ is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 5

7. 3200 in scientific notation is: 1

(a) 32×10^3 **(b)** 3.2×10^4 **(c)** 3.2×10^2 **(d)** 3.2×10^3

8. The place value of the digit 6 in 3652 is: 1

(a) 6 **(b)** 60 **(c)** 600 **(d)** 6000

9. A parallelogram has adjacent sides 8 cm and 5 cm. Its perimeter is: 1

(a) 26 cm **(b)** 21 cm **(c)** 13 cm **(d)** 40 cm

10. The sum of two odd numbers is always: 1

(a) a multiple of 3 **(b)** even **(c)** a prime number **(d)** odd

11. The sum of the first 5 odd natural numbers is: 1

(a) 10 **(b)** 30 **(c)** 24 **(d)** 25

12. 3.4×10^3 in usual form is: 1

(a) 0.0034 **(b)** 3400 **(c)** 340 **(d)** 34000

13. The digits used in the base-2 system are: 1

(a) 0 to 1 **(b)** 0 to 2 **(c)** 0 to 9 **(d)** 1 to 2

14. In a rectangle ABCD, the diagonal AC is 10 cm. The diagonal BD is: 1

(a) 20 cm **(b)** 5 cm **(c)** 10 cm **(d)** 12 cm

15. The number obtained by reversing the digits of 3472 is: 1

(a) 3481 **(b)** 2753 **(c)** 2743 **(d)** 2734

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Is 225 a perfect square? Give a reason. 2

17. Write 2^{-6} as a fraction. 2

18. Convert 111_2 to a decimal number. 2

19. One angle of a parallelogram is 48°. Find the other three angles. 2

20. Find the smallest digit to put in the blank so that the number 47_8 is divisible by 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Write 4086 in expanded form. 3

22. The diagonals AC and BD of a rectangle ABCD meet at O. If AC = 16 cm, find BD, OA and OB. 3

23. Replace * by every possible digit so that 2*41 is divisible by 3. 3

24. Write 36 as the sum of the first 6 odd natural numbers (show the sum pattern). 3

OR

Express 73000000 and 0.0000064 in scientific notation. 3

25. Convert 123_5 to a decimal number. 3

OR

The diagonals of a parallelogram ABCD meet at O with OA = 5 cm and OB = 8 cm. Find AC and BD. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Check whether 3690 is divisible by 2, 3, 5, 9 and 10. Give a reason in each case. 5

OR

Find the smallest number by which 108 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

27. Simplify $(2^2)^3 \div 2^4 \times 2^1$ and find its value. 5

OR

Convert 19 to binary by repeated division by 2 and show the steps. 5

END OF QUESTION PAPER — Total: 50 Marks